B.Sc. Engg. CSE 7th Semester | 15 April 2025

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION | **WINTER SEMESTER, 2023-2024**

DURATION: 3 HOURS | **FULL MARKS: 150**

CSE 4711: Artificial Intelligence

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses. The symbols have their usual meanings.

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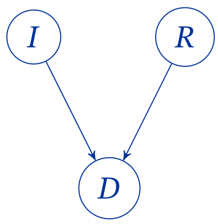
1. A contagious disease has recently broken out on your university campus. Although you are currently asymptomatic, you are concerned about the possibility of being infected (D). You have recently received a vaccination, which may provide immunity (I) against the disease. If you are immune, there is no possibility of being infected. However, there is a 20% chance that the vaccine did not successfully develop immunity, meaning there is still a chance of infection.

   The following probabilities apply under the assumption that you are not immune. You observe that your roommate (R) is feeling unwell but has not developed any specific symptoms. The chance that your roommate is infected is 40%. If your roommate is infected, the chance that you are also infected is 70%. Conversely, if your roommate is not infected, the chance that you remain uninfected is 80%.

   a) Formulate the scenario as a Bayes Net considering causal relations and construct the associated Conditional Probability Tables. 12 (CO2) (PO2)

Solution:

We have the following Bayes' Net:



The associated CPT's are as follows:

$P(I)$	
$+i$	0.8
$-i$	0.2

$P(R)$	
$+r$	0.4
$-r$	0.6

$P(D\|I,R)$			
$+i$	$+r$	$+d$	0
$+i$	$+r$	$-d$	1.0
$+i$	$-r$	$+d$	0
$+i$	$-r$	$-d$	1.0
$-i$	$+r$	$+d$	0.7
$-i$	$+r$	$-d$	0.3
$-i$	$-r$	$+d$	0.2
$-i$	$-r$	$-d$	0.8

Rubric:

- 1 point for each correct node
- 1 point for each correct edge
- 1 point for no wrong edge/node
- 0.5 points for each probability entry

b) Prove or disprove the following assumptions using algebraic notations or counterexample: 2×3 (CO1) (PO1)

i. $I \perp\!\!\!\perp R$

Solution:

We know that,

$$P(d|i,r) = \frac{P(d,i,r)}{P(i,r)}$$
$$P(d|i,r) = \frac{P(i)P(r)P(d|i,r)}{P(i,r)}$$
$$P(i,r) = P(i)P(r)$$

Hence, Immunity and Roommate's Disease are independent given no additional information.

Rubric:

- 1 point for the initial breakdown.
- 1 point for Bayes' Net implication.
- 1 point for the conclusion.

ii. $I \perp\!\!\!\perp R|D$

Solution:

If I have the disease, there is high probability of me not being immune or my roommate being affected.
If I am not immune, the probability of my roommate's disease drops and vice versa.
Hence, they are not independent.

Rubric:

- 1 point for each argument.
- 1 point for the conclusion.

c) Determine the chance of your roommate having the disease given you also have it. 8 (CO1) (PO1)

Solution:

We know:

$$P(+r|+d) = \frac{P(+r,+d)}{P(+d)}$$

Now, to determine $P(+r,+d)$ and $P(+d)$, we need to construct the joint distribution table. The joint distribution:

			$P(I,R,D)$
$+i$	$+r$	$+d$	0
$+i$	$+r$	$-d$	0.32
$+i$	$-r$	$+d$	0
$+i$	$-r$	$-d$	0.48
$-i$	$+r$	$+d$	0.056
$-i$	$+r$	$-d$	0.024
$-i$	$-r$	$+d$	0.024
$-i$	$-r$	$-d$	0.096

We get,

$$\begin{aligned} P(+r,+d) &= \sum_i P(i,+r,+d) \\ &= 0.056 \\ P(+d) &= \sum_{i,r} P(i,r,+m) \\ &= 0.080 \end{aligned}$$

So, $P(+r|+d) = \frac{0.056}{0.080} = 0.7 = 70\%$

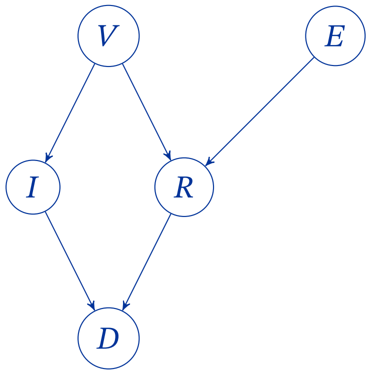
Rubric:

- 1 point for the formula.
- 0.5 points for each joint probability.
- 1 point for marginal probability.
- 1 point for the solution.

2. Building upon the scenario described in Question 1, you seek additional information to better assess your risk of infection. After discussing with your roommate, you learn that they attended an event over the weekend, where there was a possibility that another attendee was infected with the disease (E). Furthermore, both you and your roommate received vaccinations at a clinic that later reported a vaccine mix-up. Whether or not you received the correct vaccine (V) impacts your immunity status. Both the incidents also influences the likelihood that your roommate may have been contracted the diseases.

   a) Re-formulate the Bayes Net to incorporate your new observations. 11 (CO2) (PO2)

Solution:



Rubric:

- 1 point for each correct node.
- 1 point for each correct edge.
- 1 point for no wrong node/edge.

b) With a brief justification, state whether each of the following statements are guaranteed to be true for the Bayes' Net constructed in Question 2.a: 6×3 (CO1) (PO1)

i. $V \perp\!\!\!\perp D | I, R$

Solution:
Both paths $(V, I, D$ and $V, R, D)$ are inactive. Hence, guaranteed.
Rubric:

- 1 point for the statement.
- 2 points for the explanation.

ii. $V \perp\!\!\!\perp D | R$

Solution:
The path V, I, D is active. Hence, not guaranteed.
Rubric:

- 1 point for the statement.
- 2 points for the explanation.

iii. $D \perp\!\!\!\perp E | R$

Solution:
The path E, R, V, I, D is active. Hence, not guaranteed.
Rubric:

- 1 point for the statement.
- 2 points for the explanation.

iv. $V \perp\!\!\!\perp E$

Solution:
Both paths $(V, R, E$ and $V, I, D, R, E)$ are inactive. Hence, guaranteed.
Rubric:

- 1 point for the statement.
- 2 points for the explanation.

v. $V \perp\!\!\!\perp E | D$

Solution:
The path V, I, D, R, E is active. Hence, not guaranteed.
Rubric:

- 1 point for the statement.
- 2 points for the explanation.

vi. $V \perp\!\!\!\perp E | I$

Solution:
Both paths $(V, R, E$ and $V, I, D, R, E)$ are inactive. Hence, guaranteed.
Rubric:

- 1 point for the correct statement.
- 2 points for the explanation.

3. Consider an agent navigating the stochastic grid depicted in Figure 1. The agent can move in either `Up`, `Down`, `Left`, or `Right` directions unless these actions would move it off the grid or result in it hitting the wall in cell F. Such invalid actions result in no movement. Additionally, the only action available in cells H and L is `Exit`. The values in each cell indicate the random initial values of those cells chosen by the agent.

I 0.90	J 1.00	K 1.10	L 1.20
E 0.50	F	G 0.70	H 0.80
A 0.10	B 0.20	C 0.30	D 0.40

Figure 1: Grid for Question 3

a) Consider that you observe three training episodes based on some policy for your agent as shown in Figure 2. 2×11 (CO1) (PO1)

Episode 1
```
A, Up    , E, -0.04
E, Up    , I, -0.04
I, Right, E, -0.04
E, Up    , I, -0.04
I, Right, J, -0.04
J, Right, K, -0.04
K, Right, L, -0.04
L, Exit  , -, +1.00
```

Episode 2
```
A, Up    , E, -0.04
E, Up    , I, -0.04
I, Right, J, -0.04
J, Right, K, -0.04
K, Right, G, -0.04
G, Up    , K, -0.04
K, Right, L, -0.04
L, Exit  , -, +1.00
```

Episode 3
```
A, Up    , E, -0.04
E, Up    , I, -0.04
I, Right, J, -0.04
J, Right, K, -0.04
K, Right, G, -0.04
G, Up    , H, -0.04
H, Exit  , -, -1.00
```

Figure 2: Training Episodes for Agent in Question 3.a

Assume that the discount factor is 1 and the learning rate is 0.5. Now, determine the value of the states for each of the following algorithms:

i. Direct Evaluation.

Solution:

- $V(A) = \frac{0.72+0.72-1.24}{3} = 0.06$
- $V(B) = 0.20$ (or we can set it to 0)
- $V(C) = 0.30$ (or we can set it to 0)
- $V(D) = 0.40$ (or we can set it to 0)
- $V(E) = \frac{0.76+0.76-1.20}{3} = 0.11$
- $V(G) = \frac{0.92-1.04}{2} = -0.06$
- $V(H) = -1.00$
- $V(I) = \frac{0.80+0.80-1.16}{3} = 0.15$
- $V(J) = \frac{0.92+0.84-1.12}{3} = 0.21$
- $V(K) = \frac{0.96+0.88-1.08}{3} = 0.25$
- $V(L) = \frac{1+1}{2} = 1.00$

Rubric:

- 1 point for each correct value of the cells.

ii. Temporal Difference Learning

Solution:

For Episode 1:

- Sample 1: $-0.04 + 1 \times 0.50 = 0.46$
  $V(A) = (1 - 0.5) \times 0.10 + 0.5 \times 0.46 = 0.28$
- Sample 2: $-0.04 + 1 \times 0.90 = 0.86$
  $V(E) = (1 - 0.5) \times 0.50 + 0.5 \times 0.86 = 0.68$
- Sample 3: $-0.04 + 1 \times 0.68 = 0.64$
  $V(I) = (1 - 0.5) \times 0.90 + 0.5 \times 0.64 = 0.77$
- Sample 4: $-0.04 + 1 \times 0.77 = 0.73$
  $V(E) = (1 - 0.5) \times 0.68 + 0.5 \times 0.73 = 0.71$
- Sample 5: $-0.04 + 1 \times 1.00 = 0.96$
  $V(I) = (1 - 0.5) \times 0.77 + 0.5 \times 0.96 = 0.87$
- Sample 6: $-0.04 + 1 \times 1.10 = 1.06$
  $V(J) = (1 - 0.5) \times 1.00 + 0.5 \times 1.06 = 1.03$
- Sample 7: $-0.04 + 1 \times 1.20 = 1.16$
  $V(K) = (1 - 0.5) \times 1.10 + 0.5 \times 1.16 = 1.13$
- Sample 8: 1.00
  $V(L) = (1 - 0.5) \times 1.20 + 0.5 \times 1.00 = 1.10$

For Episode 2:

- Sample 1: $-0.04 + 1 \times 0.71 = 0.67$
  $V(A) = (1 - 0.5) \times 0.28 + 0.5 \times 0.67 = 0.48$
- Sample 2: $-0.04 + 1 \times 0.87 = 0.83$
  $V(E) = (1 - 0.5) \times 0.71 + 0.5 \times 0.83 = 0.77$
- Sample 3: $-0.04 + 1 \times 1.03 = 0.99$
  $V(I) = (1 - 0.5) \times 0.87 + 0.5 \times 0.99 = 0.93$
- Sample 4: $-0.04 + 1 \times 1.13 = 1.09$
  $V(J) = (1 - 0.5) \times 1.03 + 0.5 \times 1.09 = 1.06$
- Sample 5: $-0.04 + 1 \times 0.70 = 0.66$
  $V(K) = (1 - 0.5) \times 1.13 + 0.5 \times 0.66 = 0.90$
- Sample 6: $-0.04 + 1 \times 0.9 = 0.86$
  $V(G) = (1 - 0.5) \times 0.70 + 0.5 \times 0.86 = 0.78$
- Sample 7: $-0.04 + 1 \times 1.10 = 1.06$
  $V(K) = (1 - 0.5) \times 0.90 + 0.5 \times 1.06 = 0.98$
- Sample 8: 1.00
  $V(L) = (1 - 0.5) \times 1.10 + 0.5 \times 1.00 = 1.05$

For episode 3,

- Sample 1: $-0.04 + 1 \times 0.77 = 0.73$
  $V(A) = (1 - 0.5) \times 0.48 + 0.5 \times 0.73 = 0.61$

- Sample 2: $-0.04 + 1 \times 0.93 = 0.89$
  $V(E) = (1 - 0.5) \times 0.77 + 0.5 \times 0.89 = 0.83$
- Sample 3: $-0.04 + 1 \times 1.06 = 1.02$
  $V(I) = (1 - 0.5) \times 0.93 + 0.5 \times 1.02 = 0.98$
- Sample 4: $-0.04 + 1 \times 0.98 = 0.94$
  $V(J) = (1 - 0.5) \times 1.06 + 0.5 \times 0.94 = 1.00$
- Sample 5: $-0.04 + 1 \times 0.78 = 0.74$
  $V(K) = (1 - 0.5) \times 0.98 + 0.5 \times 0.74 = 0.86$
- Sample 6: $-0.04 + 1 \times 0.80 = 0.76$
  $V(G) = (1 - 0.5) \times 0.78 + 0.5 \times 0.76 = 0.77$
- Sample 7: -1.00
  $V(H) = (1 - 0.5) \times 0.80 - 0.5 \times 1.00 = -0.10$

The final values are:

- $V(A) = 0.61$
- $V(B) = 0.20$
- $V(C) = 0.30$
- $V(D) = 0.40$
- $V(E) = 0.83$
- $V(G) = 0.77$
- $V(H) = -0.10$
- $V(I) = 0.98$
- $V(J) = 1.03$
- $V(K) = 0.86$
- $V(L) = 1.05$

Rubric:

- 1 point for each correct value of the cells.

b) Between the two algorithms mentioned in Question 3.a, which one would you prefer in this scenario? Justify your choice. $1 + 2$ (CO1) (PO1)

Solution:

I would prefer Temporal Difference Learning.
Reasoning:

- The values of the states update rapidly, as a result will require less number of episodes to converge.
- It leverages the connection among states.

Rubric:

- 1 point for the correct choice.
- 1 point each for each of the reasoning.

c) Considering either set of values found in Question 3.a, is it possible to determine the optimal policy? Justify your decision. If it is possible, recommend the optimal policy for the agent. If it is not, provide an alternative. $1 + 2 + 11$ (CO3) (PO3)

Solution:

It is not possible to determine the optimal policy from the values, since policy extraction requires the transition function and the reward function, which is absent in this scenario. An alternative can be to use Q-Learning. Instead of learning the value of the states, we can learn Q-values using:

$$Q(s,a) \leftarrow (1-\alpha)Q(s,a) + \alpha \times \left[R(s,a,s') + \gamma \max_{a}^{'} Q(s',a')\right]$$

Rubric:

- 1 point for the correct decision.
- 2 points for the justification.
- 11 points for the alternative.

4. a) Some researchers argue that perception and motor skills form the core of intelligence, with higher-level abilities being secondary additions. Much of a person's brain is primarily focused on these core functions, while Artificial Intelligence has found higher-level tasks like Game Playing and Logical Reasoning easier compared to real-world perception or action. 6 (CO3) (PO3)

Comment on whether AI's emphasis on higher-level cognitive functions is misguided or not.

Solution:

Perception and motor skills are certainly important, and fields like vision and robotics are valuable (whether or not they are seen as part of "core" AI). However, once an agent receives a percept, it must still "decide" (either through deliberation or reaction) which action to take. This is as true in the real world as in artificial scenarios like chess. Therefore, determining the right action will always be essential to AI, regardless of the perceptual or motor systems to which the agent is connected to. That said, focusing too much on microworlds has distracted AI from more complex environments like those faced by self-driving cars.

Rubric:

- 3 points for justification for decision making.
- 3 points for acknowledging the lack of focus in perception and motor skills.

b) Pascal sends an n-bit message to Fermat, adding one parity bit so that the message can be recovered if at most one error occurs. Independently, each transmitted bit can be corrupted with probability p. If Pascal wants to ensure that the overall probability of Fermat getting the correct message is at least q, determine the maximum feasible value of n. 4 (CO1) (PO1)

Solution:

The correct message is received if either zero or one of the $(n+1)$ bits are corrupted.
Since corruption occurs independently, the probability that zero bits are corrupted: $(1-p)^{n+1}$.
Since there are $(n+1)$ ways of choosing one bit from the message, the probability that one bit is corrupted in a message: $\binom{n+1}{1}p(1-p)^n = (n+1)p(1-p)^n$.
The maximum feasible value of n is the largest n that satisfies the inequality:

$$(1-p)^{n+1} + (n+1)p(1-p)^n \geq q$$

Rubric:

- 1 point for the assumption.
- 1 point for each correct probability.
- 1 point for the inequality.

5. Consider an agent navigating the stochastic grid depicted in Figure 3.

S	A	
B	C	G

Figure 3: Grid for Question 5

The agent follows a set of rules while moving through the grid:

- The agent can move *Down* or *Right* unless these actions would move it off the grid.
- When moving *Right* from cell B, there is a 30% chance the agent will slip, causing it to move two cells to the right instead of one. Otherwise, it moves only one cell to the right.
- From cells S, A, and C, the agent's movement in any valid direction is deterministic, always advancing one cell.
- Upon reaching cell G, the agent can perform only the $Exit$ action, which provides a reward of 100. No further actions are available after exiting.
- Any move resulting in the agent entering cell A yields $+5$ reward. For all the other transitions, moving one cell results in a -10 reward and moving two cells results in a -4 reward.

The agent's processor is capable of handling at most two iterations of calculations for this task. Now, answer the following questions:

a) Formulate the scenario as a Markov Decision Process by identifying the set of states, the set of actions, the transition function, and the reward function. 12 (CO2) (PO2)

Solution:

States: S, A, B, C, G
Actions: Right, Down, Exit
The transition function and the reward function are given as follows:

s	a	s'	$T(s,a,s')$	$R(s,a,s')$
S	Right	A	1.0	$+5$
S	Down	B	1.0	-10
A	Down	C	1.0	-10
B	Right	C	0.7	-10
B	Right	G	0.3	-4
C	Right	G	1.0	-10
G	Exit	End	1.0	100

Rubric:

- 0.5 points for each item.
- 1 point for not including any wrong additional information.

b) Assume that the discount factor is 0.4. Recommend the optimal policy for the agent. 20 (CO3) (PO3)

Solution:

For $k = 0$, we have $V_0(S) = 0$ for $S \in \{S, A, B, C, G\}$.
For $k = 1$, we have:
For S,

$$\begin{aligned} V_1(S) = \max(&T(S, Right, A) \times [R(S, Right, A) + \gamma \times V_0(A)], \\ &T(S, Down, B) \times [R(S, Down, B) + \gamma \times V_0(B)]) \\ = \max&(1 \times [5 + 0.4 \times 0], 1 \times [-10 + 0.4 \times 0]) \\ =5& \end{aligned}$$

For A,

$$\begin{aligned} V_1(A) =& T(A, Down, C) \times [R(A, Down, C) + \gamma \times V_0(C)] \\ =& 1 \times [-10 + 0.4 \times 0] \\ =& -10 \end{aligned}$$

For B,

$$\begin{aligned} V_1(B) = &(T(B, Right, C) \times [R(B, Right, C) + \gamma \times V_0(C)]) + \\ &(T(B, Right, G) \times [R(B, Right, G) + \gamma \times V_0(G)]) \\ = &(0.7 \times [-10 + 0.4 \times 0]) + (0.3 \times [-4 + 0.4 \times 0]) \\ = &-8.2 \end{aligned}$$

For C,

$$\begin{aligned} V_1(C) =& T(C, Right, G) \times [R(C, Right, G) + \gamma \times V_0(G)] \\ =& 1 \times [-10 + 0.4 \times 0] \\ =& -10 \end{aligned}$$

For G,

$$\begin{aligned} V_1(G) =& T(G, Exit, End) \times R(G, Exit, End) \\ =& 1 \times 100 \\ =& 100 \end{aligned}$$

For $k = 2$, we have:
For S,

$$\begin{aligned} V_2(S) = \max(&T(S, Right, A) \times [R(S, Right, A) + \gamma \times V_1(A)], \\ &T(S, Down, B) \times [R(S, Down, B + \gamma \times V_1(B)]) \\ = \max&(1 \times [5 + 0.4 \times (-10)], 1 \times [-10 + 0.4 \times (-8.2)]) \\ =1& \\ \pi(S) = \text{argmax}(&T(S, Right, A) \times [R(S, Right, A) + \gamma \times V_0(A)], \\ &T(S, Down, B) \times [R(S, Down, B + \gamma \times V_0(B)]) \\ =&Right \end{aligned}$$

For A,

$$\begin{aligned} V_2(A) =& T(A, Down, C) \times [R(A, Down, C) + \gamma \times V_1(C)] \\ =& 1 \times [-10 + 0.4 \times (-10)] \\ =& -14 \\ \pi(A) =& Down \end{aligned}$$

For B,

$$\begin{aligned}V_2(B) =&(T(B,Right,C)\times[R(B,Right,C)+\gamma\times V_1(C)])+\\&T(B,Right,G)\times[R(B,Right,G)+\gamma\times V_1(G)]\\=&(0.7\times[-10+0.4\times(-10)])+(0.3\times[-4+0.4\times 100])\\=&1\\\pi(B)=&Right\end{aligned}$$

For C,

$$\begin{aligned}V_2(C)=&T(C,Right,G)\times[R(C,Right,G)+\gamma\times V_1(G)]\\=&1\times[-10+0.4\times 100]\\=&30\\\pi(C)=&Right\end{aligned}$$

For G,

$$\begin{aligned}V_2(G)=&T(G,Exit,End)\times R(G,Exit,End)\\=&1\times 100\\=&100\\\pi(G)=&Exit\end{aligned}$$

Since the agent is only able to handle at most two iterations of calculations, we stop here. The optimal policy: $\pi(S) = Right$, $\pi(A) = Down$, $\pi(B) = Right$, $\pi(C) = Right$, and $\pi(G) = Exit$.

Rubrics:

- 0.5 points for each V_0.
- 2 points for each $V_1(S)$, $V_2(S)$, $V_1(B)$, and $V_2(B)$.
- 1 point for each $V_1(A)$, $V_2(A)$, $V_1(C)$, $V_2(C)$, $V_1(G)$, $V_2(G)$.
- 1 point for acknowledging the limit.
- 0.5 points for each optimal action.

6. Assume that your wealth is $\$k$. You are considering investing \$1 in a newly launched cryptocurrency, Dodgecoin. This investment carries significant risk, but it also offers potential rewards. There are two possible positive outcomes:

- A \$10 return with probability 1/50
- A $\$1,000,000$ return with probability 1/2000000

Additionally, in both cases, you will get your investment back. However, there is also a high probability that the investment will result in no gains.

a) Determine the expected monetary value of the investment. 2 (CO1) (PO1)

Solution:

$EMV(C) = \frac{1}{50}\times(\$10+\$1)+\frac{1}{2000000}\times(\$1000000+\$1)\approx \$0.72$

Rubric:

- 1 point for the equation
- 1 point for the result

b) Assess under what conditions it would be rational to invest in Dodgecoin. 4 (CO1) (PO1)

Solution:

To formulate when it is rational to invest in the cryptocurrency, we need to compare the expected utility of investing \$1 with the utility of keeping that \$1.
The expected utility of investing is:

$$EU_{\text{invest}} = \left(\frac{1}{50}\right) U(\$k{+}\$10){+}\left(\frac{1}{2000000}\right) U(\$k{+}\$2000000){+}\left(1 - \frac{1}{50} - \frac{1}{2000000}\right) U(\$k{-}\$1)$$

If we do not invest, our wealth remains k, so the expected utility is: $EU_{\text{no invest}} = U(\$k)$.
For investing to be rational:

$$\left(\frac{1}{50}\right) U(\$k{+}\$10){+}\left(\frac{1}{2000000}\right) U(\$k{+}\$2000000){+}\left(1 - \frac{1}{50} - \frac{1}{2000000}\right) U(\$k{-}\$1) \geq U(\$k)$$

Rubric:

- 2 point for EU_{invest}
- 1 point for $EU_{\text{no invest}}$
- 1 point for the condition

c) Sociological studies show that individuals with lower income are more likely to invest a disproportionate amount in highly speculative assets like cryptocurrencies. Evaluate whether this behavior results from poorer decision-making or a difference in the utility function. 8 (CO2) (PO2)

Solution:

There are several factors that could explain why individuals with lower incomes disproportionately invest in cryptocurrency:

- Some investors may not fully account for the low probability of a large return and may instead focus on the magnitude of the potential payoff. This leads them to overestimate the attractiveness of the investment, making them prone to irrational decision-making. They may place undue weight on the possibility of a large win, disregarding the very low chances of it actually occurring.

- Individuals in difficult financial situations may perceive the difference between having one dollar more or less as negligible. For them, the risk of losing a small amount of money may not seem significant compared to the potential large payoff. This makes it rational, from a utility perspective, to take the gamble. Essentially, the expected loss is small enough that they are willing to take the chance for the possibility of a significant gain, which might outweigh the downside risk in their subjective valuation of wealth.

- Another possibility is that some investors may derive utility from the excitement and entertainment value of the investment itself, rather than solely focusing on monetary returns. For these individuals, purchasing cryptocurrency could be seen as akin to buying a movie ticket – the thrill and excitement of participation, even if the expected outcome is a loss, justify the cost. In this case, the decision is not purely economic but emotional or recreational.

The disproportionate investment in cryptocurrency by lower-income individuals can likely be attributed to a combination of irrational decision-making, a different perception of risk and reward due to financial constraints, and the emotional or entertainment value

derived from the investment. These factors suggest that the behavior may stem from both different utility functions and suboptimal decision-making rather than solely from poor judgment.
Rubric:

- 2 point for each reason
- 2 point for conclusion